

Bash scripting task 1

Bash scripting : a file that contains linux commands
instead of typing commands one by one we can use write all
commands in one script file

- * Echo \$0 tells use finding the current shell
- * cat /etc/shells for knowing shells in our o.s

```
Last login: Wed Nov  5 13:33:52 2025 from 13.48.4.203
[ec2-user@ip-172-31-4-203 ~]$ echo $0
-bash
[ec2-user@ip-172-31-4-203 ~]$ cat /etc/shells
/bin/sh
/bin/bash
/usr/bin/sh
/usr/bin/bash
/bin/csh
/bin/tcsh
/usr/bin/csh
/usr/bin/tcsh
```

Sh (it is an shell which will open) same with other shells like csh

```
[ec2-user@ip-172-31-4-203 ~]$ sh
sh-5.2$ echo $0
sh
sh-5.2$ exit
exit
[ec2-user@ip-172-31-4-203 ~]$ csh
[ec2-user@ip-172-31-4-203 ~]$ echo $0
csh
[ec2-user@ip-172-31-4-203 ~]$ exit
exit
```

basic Shell Scripts:

=====

Create a basic script with output: Hello World!

Basic Second script:

=====

#!/bin/bash

pwd

ls

whoami

date

cal

touch a b c

```
#!/bin/bash
#purpose: for testing script format
#date: 06-11-2025
#modification: 06-11-2025
pwd
ls
whoami
date
cal
touch a b c
```

```
~
~
```

Chmod 755 for giving permission to file

```
[root@ip-172-31-4-203 scripts]# chmod 755 helloworld
[root@ip-172-31-4-203 scripts]# cd
```

For running bash commands we use

- 1) ./helloworld.bash
- 2) Bash helloworld.bash (runs file without permission)

```
[[root@ip-172-31-4-203 scripts]# vi helloworld.bash
[[root@ip-172-31-4-203 scripts]# ./helloworld.bash
/root/waseem/scripts
abc.bash  helloworld.bash
root
Thu Nov  6 10:09:47 UTC 2025
    November 2025
Su Mo Tu We Th Fr Sa
                1
 2  3  4  5  6  7  8
 9 10 11 12 13 14 15
16 17 18 19 20 21 22
23 24 25 26 27 28 29
30
[[root@ip-172-31-4-203 scripts]# ls
a  abc.bash  b  c  helloworld.bash
[[root@ip-172-31-4-203 scripts]#
```

We can use —> bash helloworld.bash for ignoring the permissions

```
[[root@ip-172-31-4-203 scripts]# vi abc
[[root@ip-172-31-4-203 scripts]# ll
total 8
-rw-r--r--. 1 root root 38 Nov  6 09:50 abc
-rwxr-xr-x. 1 root root 29 Nov  6 09:27 helloworld.bash
[[root@ip-172-31-4-203 scripts]# ./abc
-bash: ./abc: Permission denied
[[root@ip-172-31-4-203 scripts]# bash abc
hello this is waseem
[[root@ip-172-31-4-203 scripts]#
```

Script format :~

```
#!/bin/bash
```

```
# Purpose: Testing script format
```

```
#Date: 06-11-2025
```

```
#Modification: 06-11-2025
```

```
a='hello world'
```

```
echo $a
```

```
#!/bin/bash
#purpose: for testing script format
#date: 06-11-2025
#modification: 06-11-2025

a='hello world'

echo $a
```

./helloworld.bash

Cat helloworld.bash for viewing content

```
[[root@ip-172-31-4-203 scripts]# vi helloworld.bash
[[root@ip-172-31-4-203 scripts]# ./helloworld.bash
hello world
[[root@ip-172-31-4-203 scripts]# cat helloworld.bash

#!/bin/bash
#purpose: for testing script format
#date: 06-11-2025
#modification: 06-11-2025

a='hello world'

echo $a
[[root@ip-172-31-4-203 scripts]#
```

Basic Administration Tasks:

```
#!/bin/bash
top | head -10
df -h
free -m
uptime
iostat
```

```
[root@ip-172-31-4-203 scripts]# vi user.bash
[root@ip-172-31-4-203 scripts]# cat user.bash
#!/bin/bash
top | head -10
df -h
free -m
uptime
iostat
[root@ip-172-31-4-203 scripts]# ./user.bash

top - 10:18:25 up 1:09, 2 users, load average: 0.00, 0.00, 0.00
Tasks: 104 total, 1 running, 103 sleeping, 0 stopped, 0 zombie
%Cpu(s): 3.2 us, 0.0 sy, 0.0 ni, 96.8 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 904.8 total, 502.8 free, 177.6 used, 224.4 buff/cache
MiB Swap: 0.0 total, 0.0 free, 0.0 used. 597.8 avail Mem

  PID USER      PR  NI   VIRT   RES   SHR  S  %CPU  %MEM     TIME+ COMMAND
    21 root        20   0      0      0      0   S   6.7   0.0   0:00.05 ksoftirqd/1
     1 root        20   0 172248 17312 10836   S   0.0   1.9   0:01.11 systemd
     2 root        20   0      0      0      0   S   0.0   0.0   0:00.00 kthreadd

Filesystem            Size  Used Avail Use% Mounted on
devtmpfs               4.0M    0  4.0M   0% /dev
tmpfs                  453M    0  453M   0% /dev/shm
tmpfs                  181M  428K  181M   1% /run
/dev/nvme0n1p1         8.0G  1.7G   6.4G  21% /
tmpfs                  453M    0  453M   0% /tmp
/dev/nvme0n1p128       10M  1.3M   8.7M  13% /boot/efi
tmpfs                  91M    0   91M   0% /run/user/1000
      total            used        free      shared  buff/cache   available
Mem:           904         177         502           0         224         597
Swap:           0           0           0

 10:18:25 up 1:09, 2 users, load average: 0.00, 0.00, 0.00
Linux 6.1.156-177.286.amzn2023.x86_64 (ip-172-31-4-203.eu-north-1.compute.internal) 11/06/25 _x86_64_ (2 CPU)

avg-cpu:  %user   %nice %system %iowait  %steal   %idle
           0.09    0.00    0.07    0.03    0.02   99.78

Device            tps    kB_read/s    kB_wrtn/s    kB_dscd/s    kB_read    kB_wrtn    kB_dscd
nvme0n1            2.55        53.11        21.46         0.00     221379     89472         0

[root@ip-172-31-4-203 scripts]#
```

Cat user.bash for content display
./user.bash for running script

Defining Variables:

In programming, a variable is a value that can change, depending on conditions or on information passed to the program.

Example:

```
#!/bin/bash
#Define Variables
p=pwd
l=ls
wi=whoami
d=date
c='cal 2025'
```

echo Run Variables tasks

echo

\$p

\$l

\$wi

\$d

\$c

```
#!/bin/bash
#define variable
p=pwd
l=ls
wi=whoami
d=date
c='cal 2025'

echo run variable tasks
echo
$p
$l
$wi
$d
$c
~
~
~
```

```

[root@ip-172-31-4-203 scripts]# vi variable.bash
[root@ip-172-31-4-203 scripts]# chmod 755 variable.bash
[root@ip-172-31-4-203 scripts]# ./variable.bash
run variable tasks

/root/waseem/scripts
a abc.bash b c helloworld.bash user.bash variable.bash
root
Thu Nov  6 10:28:43 UTC 2025

2025

January
Su Mo Tu We Th Fr Sa
      1  2  3  4
  5  6  7  8  9 10 11
12 13 14 15 16 17 18
19 20 21 22 23 24 25
26 27 28 29 30 31

February
Su Mo Tu We Th Fr Sa
      1
  2  3  4  5  6  7  8
 9 10 11 12 13 14 15
16 17 18 19 20 21 22
23 24 25 26 27 28

March
Su Mo Tu We Th Fr Sa
      1
  2  3  4  5  6  7  8
 9 10 11 12 13 14 15
16 17 18 19 20 21 22
23 24 25 26 27 28 29
30 31

April
Su Mo Tu We Th Fr Sa
      1  2  3  4  5
  6  7  8  9 10 11 12
13 14 15 16 17 18 19
20 21 22 23 24 25 26
27 28 29 30

May
Su Mo Tu We Th Fr Sa
      1  2  3
  4  5  6  7  8  9 10
11 12 13 14 15 16 17
18 19 20 21 22 23 24
25 26 27 28 29 30 31

June
Su Mo Tu We Th Fr Sa
  1  2  3  4  5  6  7
 8  9 10 11 12 13 14
15 16 17 18 19 20 21
22 23 24 25 26 27 28
29 30

July
Su Mo Tu We Th Fr Sa
      1  2  3  4  5
  6  7  8  9 10 11 12
13 14 15 16 17 18 19
20 21 22 23 24 25 26
27 28 29 30 31

August
Su Mo Tu We Th Fr Sa
      1  2
  3  4  5  6  7  8  9
10 11 12 13 14 15 16
17 18 19 20 21 22 23
24 25 26 27 28 29 30
31

September
Su Mo Tu We Th Fr Sa
      1  2  3  4  5  6
  7  8  9 10 11 12 13
14 15 16 17 18 19 20
21 22 23 24 25 26 27
28 29 30

October
Su Mo Tu We Th Fr Sa
      1  2  3  4
  5  6  7  8  9 10 11
12 13 14 15 16 17 18
19 20 21 22 23 24 25
26 27 28 29 30 31

November
Su Mo Tu We Th Fr Sa
      1
  2  3  4  5  6  7  8
 9 10 11 12 13 14 15
16 17 18 19 20 21 22
23 24 25 26 27 28 29
30

December
Su Mo Tu We Th Fr Sa
      1  2  3  4  5  6
  7  8  9 10 11 12 13
14 15 16 17 18 19 20
21 22 23 24 25 26 27
28 29 30 31

[root@ip-172-31-4-203 scripts]# █

```

Chmod 755 variable.bash (for permission)
 ./variable.bash (runs the script)

Input/Output script:

read --> To take command from user

echo --> to print the output

Ex1:

```
#!/bin/bash
```

```
echo Hello, my name is waseem
```

```
echo
```

```
echo what is your name?
```

```
read name
```

```
echo
```

```
echo Hello $name
```

```
echo
```

```
#!/bin/bash
echo hello.my name is waseem
echo
echo what is your name?
read name
echo hello $name
echo
```

```
[root@ip-172-31-4-203 scripts]# chmod 755 input.bash
[root@ip-172-31-4-203 scripts]# ./input.bash
hello.my name is waseem

what is your name?
```

We given name akram and entered (out put hello akram)

```
-bash: ./input.bash: Permission denied
[root@ip-172-31-4-203 scripts]# chmod 755 input.bash
[root@ip-172-31-4-203 scripts]# ./input.bash
hello.my name is waseem

what is your name?
akram
hello akram

[root@ip-172-31-4-203 scripts]#
```


Ex2:

```
#!/bin/bash
```

#we need to use the ticks if you want to use this in echo or else not required

```
a=`hostname`
```

```
echo Hello, my name is $a
```

```
echo
```

```
echo what is your name?
```

```
read b
```

```
echo
```

```
echo Hello $b
```

```
echo
```

```
#!/bin/bash
#we need to use the ticks if you want to use in echo or else not required
a=`hostname`
echo hello my name is $a
echo
echo what is your name?
read b
echo
echo hello$b
echo
```

```
~
~
~
~
~
```

```
[[root@ip-172-31-4-203 scripts]# vi output.bash
[[root@ip-172-31-4-203 scripts]# chmod 755 output.bash
[[root@ip-172-31-4-203 scripts]# ./output.bash
hello my name is hostname

what is your name?
[waseem

hellowaseem

[[root@ip-172-31-4-203 scripts]# █
```


